IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

EEE PART III/IV: MEng, BEng and ACGI

OPTOELECTRONICS

Thursday, 14 December 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : R.R.A. Syms
Second Marker(s) : K. Fobelets

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Fundamental constants

$e = 1.6 \times 10^{-19}$ C

$m_e = 9.1 \times 10^{-31}$ kg

$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

$\mu_0 = 4\pi \times 10^{-7}$ m kg/C^2

$c = 3 \times 10^8$ m/s

$h = 6.62 \times 10^{-34}$ Js

Maxwell's equations – integral form, all fields time varying

$\iint_A \mathbf{D} \cdot d\underline{a} = \iiint_V \rho \, dv$

$\iint_A \mathbf{B} \cdot d\underline{a} = 0$

$\int_L \mathbf{E} \cdot d\underline{L} = -\iint_A \partial\mathbf{B}/\partial t \cdot d\underline{a}$

$\int_L \mathbf{H} \cdot d\underline{L} = \iint_A [\mathbf{J} + \partial\mathbf{D}/\partial t] \cdot d\underline{a}$

Maxwell's equations – differential form, all fields time varying

$\mathrm{div}(\mathbf{D}) = \rho$

$\mathrm{div}(\mathbf{B}) = 0$

$\mathrm{curl}(\mathbf{E}) = -\partial\mathbf{B}/\partial t$

$\mathrm{curl}(\mathbf{H}) = \mathbf{J} + \partial\mathbf{D}/\partial t$

Material or constitutive equations

$\mathbf{J} = \sigma \mathbf{E}$

$\mathbf{D} = \varepsilon \mathbf{E}$

$\mathbf{B} = \mu \mathbf{H}$

Vector calculus (Cartesian co-ordinates)

$\mathrm{grad}(\phi) = \partial\phi/\partial x \, \underline{i} + \partial\phi/\partial y \, \underline{j} + \partial\phi/\partial z \, \underline{k}$

$\mathrm{div}(\mathbf{F}) = \partial\mathbf{F}_x/\partial x + \partial\mathbf{F}_y/\partial y + \partial\mathbf{F}_z/\partial z$

$\mathrm{curl}(\mathbf{F}) = \underline{i} \{\partial\mathbf{F}_z/\partial y - \partial\mathbf{F}_y/\partial z\} + \underline{j} \{\partial\mathbf{F}_x/\partial z - \partial\mathbf{F}_z/\partial x\} + \underline{k} \{\partial\mathbf{F}_y/\partial x - \partial\mathbf{F}_x/\partial y\}$

$\mathrm{curl}\{\mathrm{curl}(\mathbf{F})\} = \mathrm{grad}\{\mathrm{div}(\mathbf{F})\} - \nabla^2\mathbf{F}$

$\iint_A \mathbf{F} \cdot d\underline{a} = \iiint_V \mathrm{div}(\mathbf{F}) \, dv$ (Gauss' Theorem)

$\int_L \mathbf{F} \cdot d\underline{L} = \iint_A \mathrm{curl}(\mathbf{F}) \cdot d\underline{a}$ (Stokes' Theorem)

1. a) Using Stokes' theorem, show that the integral forms of Faraday's Law and Ampère's Law can be written in time-dependent form as:

$$\text{curl}(\underline{\mathbf{E}}) = -\partial\underline{\mathbf{B}}/\partial t$$
$$\text{curl}(\underline{\mathbf{H}}) = \underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t$$

[5]

b) Making suitable assumptions about the properties of optical media, derive time-independent curl expressions for optical waves.

[5]

c) Making one other assumption, combine your two curl expressions to obtain the time-independent vector wave equation. What assumption did you have to make?

[4]

d) The electric field of a particular plane electromagnetic wave can be written in time-dependent form as:

$$E = E_0 \exp(-jkz)\,\underline{j}$$

Find the divergence and curl of this field. What do your results imply about i) the wave equation and ii) the nature of electromagnetic waves?

[6]

2. a) Sketch a ray diagram for plane wave incidence on a boundary between two dielectrics, without total internal reflection. How does this diagram modify when total internal reflection occurs? Sketch and describe the transverse field in this case.

[6]

b) State Alhazen's and Snell's laws. Explain the concept of 'critical angle' and define its value. For a glass-air interface, the critical angle is 41.83°. What is the refractive index of glass? Estimate the power reflection coefficient at normal incidence.

[6]

c) A 2D parabolic index medium can be described by the refractive index variation:

$$n(x) = n_0 \sqrt{\{1 - (x/x_0)^2\}}$$

Here, n_0 and x_0 are constants, and x is the transverse distance from the axis of propagation. Show that the equation for the ray trajectory of a mode with propagation constant β travelling in the z-direction is:

$$\beta^2(dx/dz)^2 = k_0^2 n_0^2 - \beta^2 - k_0^2 n_0^2 (x/x_0)^2$$

Show also that the trajectory $x = A \sin(Bz + \psi)$ is a solution, and explain the difference between this solution and the ray trajectory in a step-index waveguide.

[8]

3. a) Describe and sketch the layout of a Y-junction-based Mach-Zehnder interferometer.

[4]

b) Explain in detail how the device can act as an amplitude modulator.

[6]

c) Figure 1 shows a single mode channel waveguide device formed from three Y-junctions. Explaining your reasoning, find input mode amplitudes at ports 3, 4 and 5 that yield outputs from ports 1 and 2 that do not involve radiation loss.

[8]

d) Show that the device does indeed conserve power under these conditions.

[2]

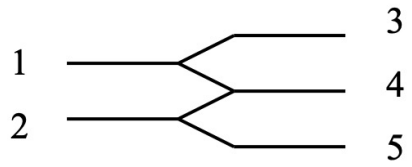


Figure 1.

4. a) Explain why titanium-diffused lithium niobate phase modulators are preferred over possible alternatives (carrier injection and electro-optic semiconductor modulators, and direct modulation of a laser) in fibre communication systems.

[4]

b) Sketch the cross-section of a Ti:$LiNbO_3$ electro-optic modulator. Indicate the location of the waveguide used with i) horizontal and ii) vertical electric fields.

[6]

c) The change in effective index induced by a voltage V using an electro-optic phase modulator formed in a material of refractive index n and with electrode gap g is:

$$\Delta n_{eff} = -n^3 r V \Gamma / 2g$$

What is the meaning of the terms r and Γ? Find the voltage-length product needed to induce a phase shift of π radians at a wavelength λ.

[6]

d) Assuming that n = 2.2, r = 30.8 x 10^{-12} m/V, λ = 1.5 μm, g = 7 μm and Γ = 0.6, calculate the voltage-length product. Hence find the voltages needed to obtain a phase shift of π radians in devices of length i) 5 mm and ii) 10 mm.

[4]

5. a) Figure 2 shows a photoconductive detector. How does it operate?

[4]

b) Estimate the responsivity of a photoconductive detector with quantum efficiency 0.6 at a wavelength of 0.88 μm.

[6]

c) List the drawbacks of photoconductive detectors for optical communications.

[4]

d) Explain the operation of an image sensor based on an array of charge coupled devices. Why do the drawbacks of photoconductive detectors matter less in this case?

[6]

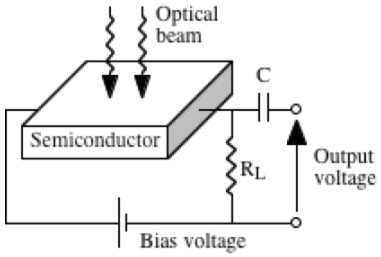


Figure 2.

Please Turn Over

6. a) The rate equations for a semiconductor laser can be written in the form:

$$dn/dt = I/ev - n/\tau_e - G\phi(n - n_0)$$
$$d\phi/dt = \beta n/\tau_{rr} + G\phi(n - n_0) - \phi/t_p$$

Explain the physical processes described by the six blocks of terms on the right-hand sides of these equations. Why is there no rate equation for holes?

[7]

b) Stating your assumptions, find an approximate steady state solution for the output power above threshold at a wavelength λ. What is the value of the slope efficiency, if the energy gap of the semiconductor material (in electron volts) is E?

[6]

c) A student's placement at **LightFantastic.com** has not gone well. They have been measuring laser diodes, which pass inspection only if the threshold current is below 13 mA and the slope efficiency is above 1.3 mW/mA. After an unfortunate incident with a fire extinguisher, all that can be read from their logbook is the following:

	I_1 (mA)	P_1 (mW)	I_2 (mA)	P_2 (mW)
Laser 1	12	1.0	20	9.0
Laser 2	18	2.8	25	12.8
Laser 3	18	8.5	24	17.0

Which of the lasers is saleable, and why?

[7]